



Mathieu Isoard

PhD | ML/AI Engineer



+ 33 6 10 69 20 15



isoard.mathieu@gmail.com



misoard.github.io

linkedin.com/in/mathieu-isoard-ml

Certifications

2024 – XCS224W Machine learning with graphs
Stanford Online

Technical Skills

LLM APIs, PyTorch, RAG, embeddings
Python, FastAPI, Docker, Supabase, build & release (Google Play, App Store)
Scikit-learn, NumPy/Pandas, TensorFlow/Keras, PyTorch

Languages

French (C2)

English (C1)

German (B1)

ML/AI engineer and applied-AI builder with a PhD in theoretical physics and applied mathematics. I design, build, and ship agentic LLM systems in production: multi-component pipelines and retrieval-grounded generation. My work spans clinical/medical imaging ML and a solo-built AI product live on both app stores, combining academic rigor with production engineering.

Experience

2026 – Present

Founder & ML Engineer @ CalenFit

- Designed and shipped an agentic two-LLM pipeline in production to turn unstructured free-text workout notes into structured data - an LLM-powered product feature live on Google Play and the App Store: calenfit.com.
- Engineered reliability & fallback: unified LLM client with retry, malformed-output detection, and model fallback; human-in-the-loop validation ensuring data integrity while hiding inference latency by turning the unavoidable wait into a UX step.
- Designed a structured-claim RAG engine grounding AI recommendations in peer-reviewed sport-science literature.

2025

Senior AI Researcher @ CRIION (Intelligent Oncology)

- Clinical/medical imaging ML: built a domain-adaptation pipeline for single-cell microscopy in oncology, combining foundation-model embeddings, Optimal Transport alignment, and ensemble voting for a +20% accuracy gain across heterogeneous datasets (different instruments, labs, and sample conditions).
- Shipped the full pipeline as a deployable web app (Python · FastAPI · React · Docker) so non-ML clinical researchers could run the methodology themselves.

2022 - 2024

Post-doc @ LKB, Sorbonne Université – Quantum Computing Group

Designed a deep neural network to detect quantum resources from sparse experimental data, outperforming existing data-hungry methods.

Education

2017 - 2020, Paris-Saclay University

PhD in theoretical Physics and applied Maths

My PhD focused on (i) uncovering quantum effects in hydrodynamic analog black holes using noisy experimental data, and (ii) solving and numerically simulating fundamental hydrodynamic equations, including shock-wave dynamics.

2013 - 2017, ENS Paris-Saclay and ENS Ulm

Bachelor (ENS Paris-Saclay) and Master degree (ENS Ulm, ICFP program)

Skills & Areas of expertise

- Agentic & multi-LLM systems · RAG & retrieval-grounded generation · prompt & context engineering
- End-to-end ML deployment · Production reliability · latency & cost optimization
- ML foundations · Project management · technical communications

Interests & Personal projects

- Sport (weightlifting and running) became a must in my life where I can apply my core beliefs to achieve success: motivation and self-discipline.
- Design a ready-to-use wellness platform for my family: equilibrecorpsspirit.com